Onboarding Powerhouse

AI Enablement in the Super Portal

Feature proposal — four AI use cases for credit decisioning

Appro Onboarding Solutions FZ-LLC

Product Owner · 21 October 2026

The Mandate — Four AI Use Cases

Agreed 20 Oct — AI embeds into existing SMBP modules, not a separate AI product. Each use case needs exactly one narrow LLM task; everything else stays deterministic.

1 · Policy Copilot

PROMPT → RULE

Credit Decision › Rule Engine (Strategies)

- Describe a policy change in plain language; pick product + time range.
- System re-runs past applications (or specific IDs) and shows how many would pass under the new policy.
- Lands as a Maker request with an AI-generated checklist; shadow mode before production.

2 · AI Reports

PROMPT → SQL

Enquiry › Report Enquiry

- Ask for a report in plain language; the system queries the database.
- Result shows the generated SQL plus a chart or diagram.
- Cathal: expose the same capability to banks through MCP.

3 · AI Document Validation

DOC COMPARISON

Document Management › Document Set Up

- Validate documents in the new Document setup module.
- Security cheques, source-of-income proofs, clearance letters checked against application data before ops review.

4 · FEEL Assist

PROMPT → FEEL

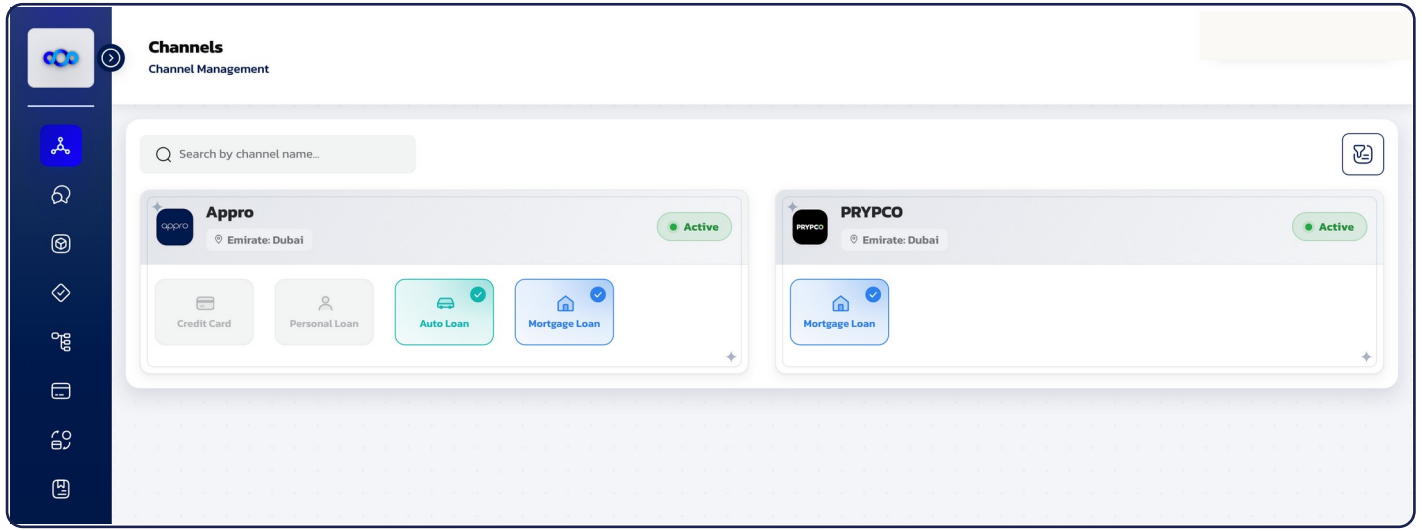
Limit Assignment › Calculated Variables · RE variables

- Write FEEL formulas from a prompt — configuration is too difficult for users today (Anto).
- Also covers use case 5: creating new Rule Engine variables with FEEL expressions.

Technical direction (Antofelix, 20 Oct): one offline, lightweight model from the Foundry catalog, specialised for these tasks, hosted in the UAE — plan and dates tracked per activity.

Design Principle — AI Drafts, Maker-Checker Decides

Every use case follows one pattern — AI drafts, the platform validates, Maker-Checker approves. No AI output reaches production unreviewed.



SMBP Super Portal — Channel Management (mirror build, demo data). All four features land inside existing modules.

The pattern — every use case, same controls

Prompt → AI draft → Validation → Maker request → Checker approval → Publish

The model proposes; parsers, replays and field-matching verify; people approve.

Why it passes review — CBUAE

Guidance Note, Feb 2026: responsible AI/ML for licensed financial institutions — human oversight on high-impact decisions, validation before deployment, drift monitoring and periodic bias testing.

Maker-checker meets this by construction — AI never holds the pen on a production decision.

Why SMBP is ready

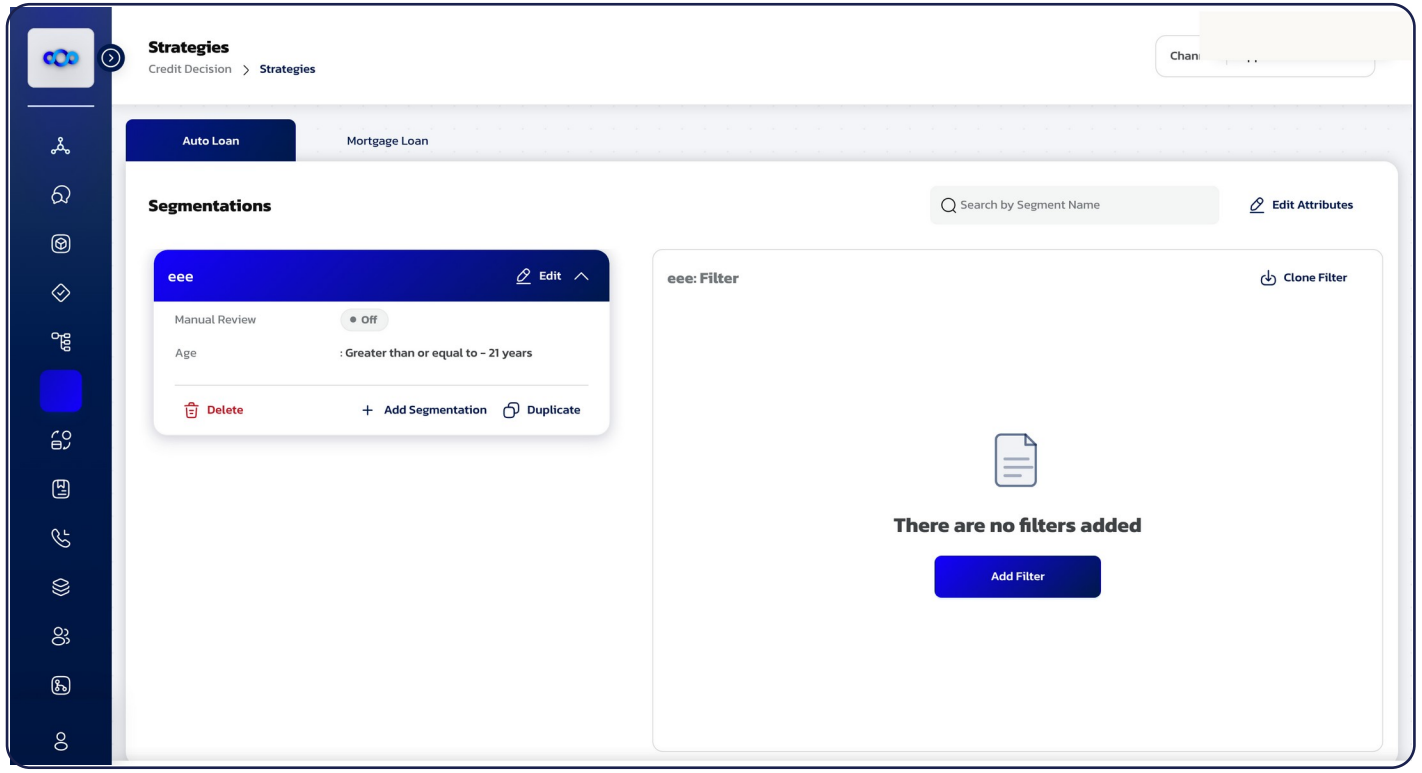
Controls already live: maker-checker on every configuration change, ~97-attribute Rule Engine catalog, formula builder and per-product document profiles.

AI plugs into controls we already run — no new governance framework to invent.

Sources: CBUAE Guidance Note on AI/ML (Feb 2026) · SMBP portal mirror (Appro build)

Policy Copilot — Write Policy, Prove It On Our Own Book

Write policy in plain language — and prove it on our own book before it touches production.



Credit Decision › Strategies (Rule Engine) — segmentations and filters: the surface Policy Copilot writes to.

Requirement · prompt → rule (20 Oct)

Back-test: prompt + product + time range → re-run past applications; show how many would pass under the new policy. Runs on specific application IDs too.

Governed: output is a Maker request with an AI checklist; shadow mode before production.

Market base

Taktile: AI copilot writes & debugs decision logic (Monzo, Allianz, Rakuten Bank; $110M Series C, 2026).

Experian Assistant: vendor-claimed model build cut from months to days.

Back-test + shadow: standard champion-challenger discipline in decisioning platforms.

Feasibility & gate

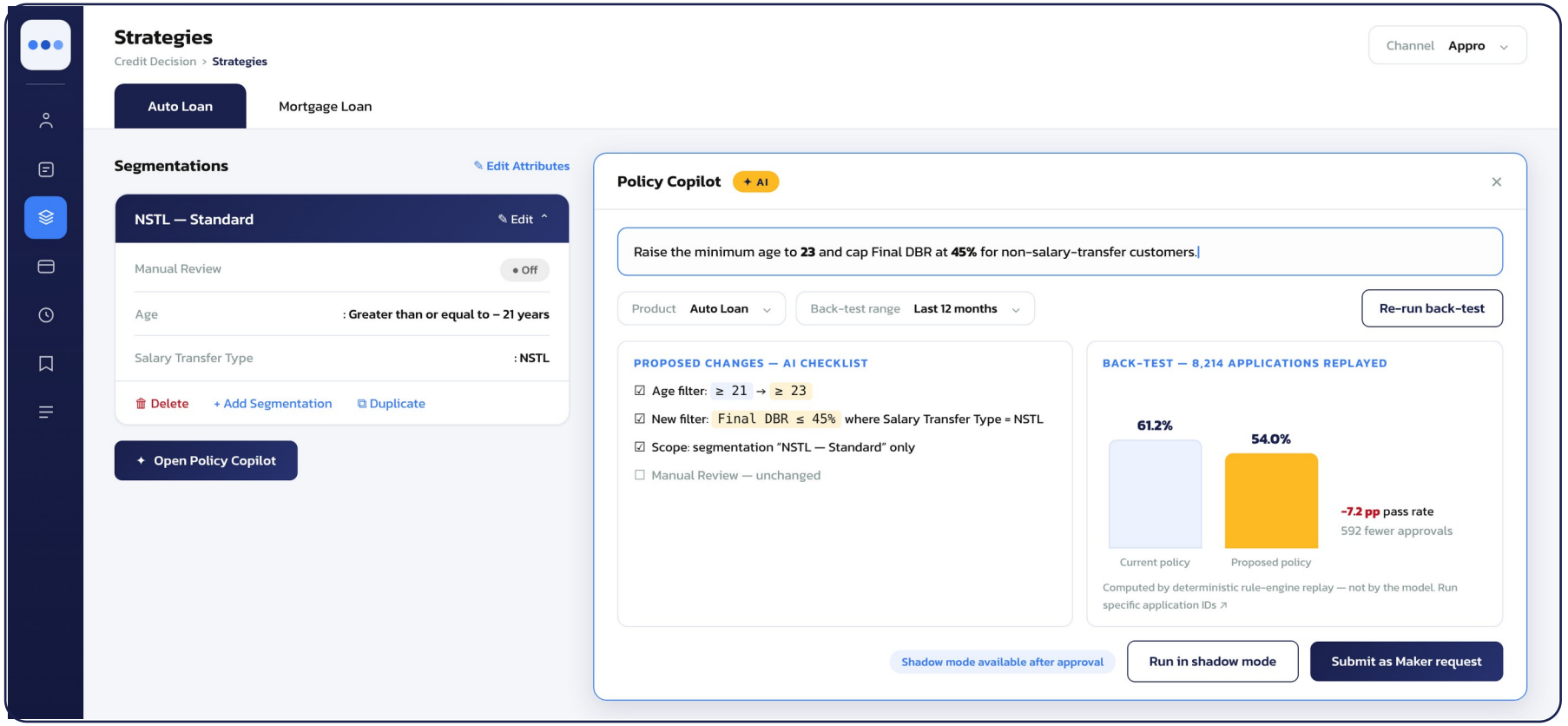
Narrow LLM task: prompt → rule JSON against our attribute catalog. Impact numbers come from deterministic replay — never from the model.

Gate: generated rules match analyst intent on a pre-agreed prompt set, then shadow pilot on one product.

Sources: Taktile (company; Business Wire 2026) · Experian (vendor claim) · Forrester AI Decisioning Wave Q2 2025 (FICO Leader)

Policy Copilot — How It Looks

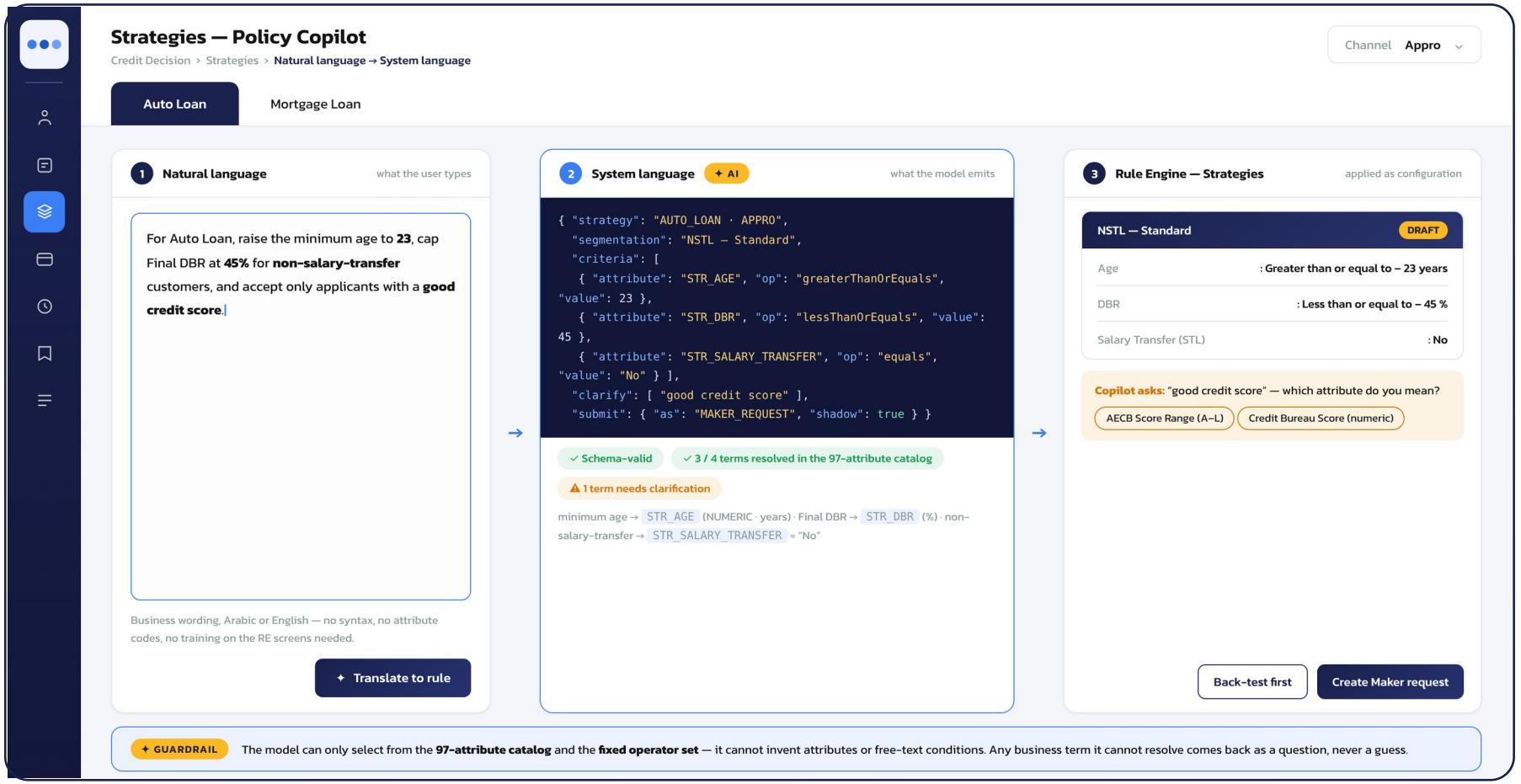
ILLUSTRATIVE MOCKUP — SAMPLE DATA



Copilot drawer in Credit Decision › Strategies — plain-language prompt, AI checklist of the proposed changes, deterministic back-test on past applications, then submission as a Maker request.

Policy Copilot — Natural Language To System Language

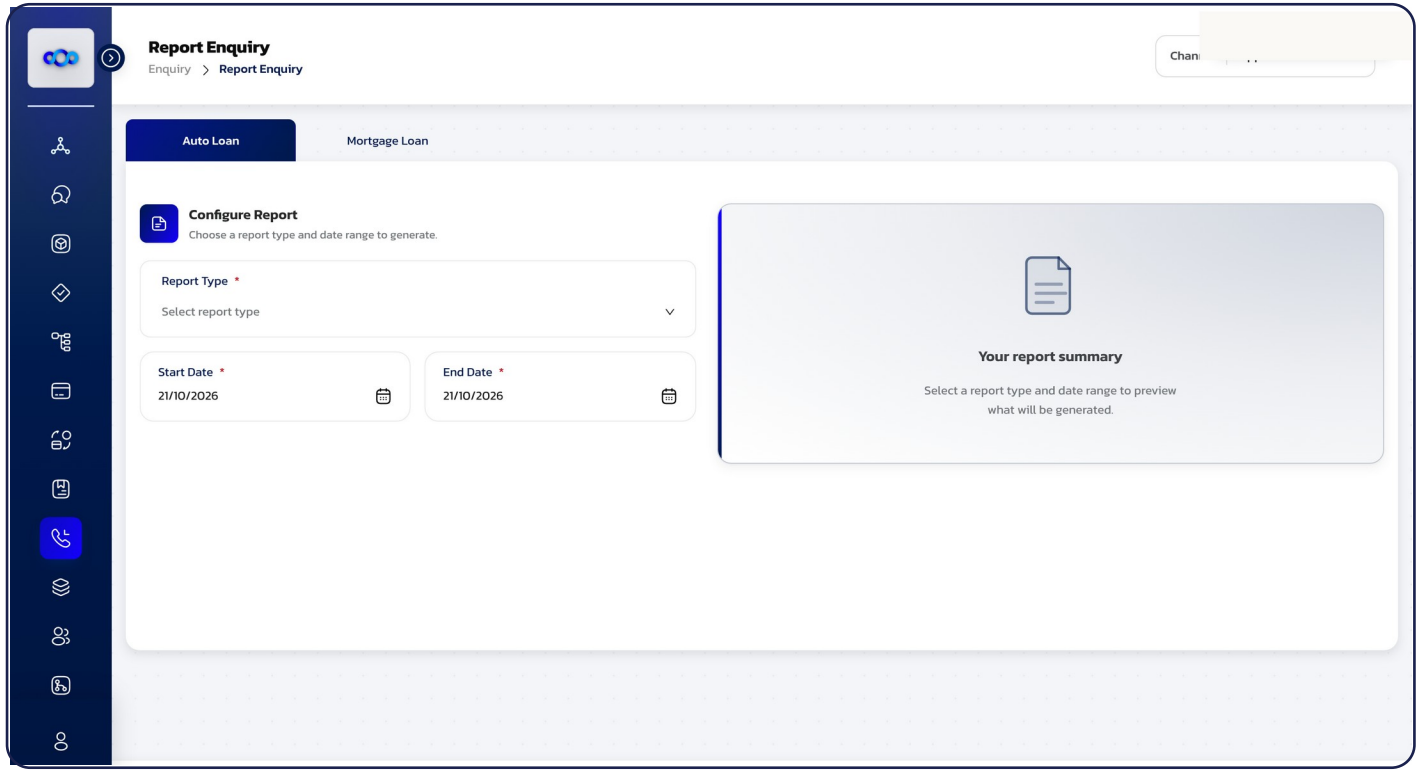
ILLUSTRATIVE MOCKUP — SAMPLE DATA



The translation step: business wording in — a schema-bound rule out, grounded in the Rule Engine's own catalog (STR_AGE, STR_DBR, STR_SALARY_TRANSFER) and fixed operator set. Unresolved terms come back as questions, never guesses.

AI Reports — Governed Semantic Layer, Not Raw NL-to-SQL

Prompt-to-report is feasible — if we constrain it to a governed semantic layer. Raw NL-to-SQL fails on enterprise schemas.



Enquiry › Report Enquiry — today: pick a report type and date range. Target: ask in plain language, see SQL + chart.

Requirement · prompt → SQL (20 Oct)

Ask in plain language; the system queries the database and returns the generated SQL with a visualization.

MCP: Cathal — expose the capability to the bank through MCP.

Market reality check

≈80% on academic leaderboards (BIRD) — but on real enterprise warehouses:

10.8% execution accuracy (BEAVER, GPT-5.2 agents)

21.3% on Spider 2.0 vs 91.2% on Spider 1.0. The schema is the bottleneck, not the model.

Hypothesis & gate

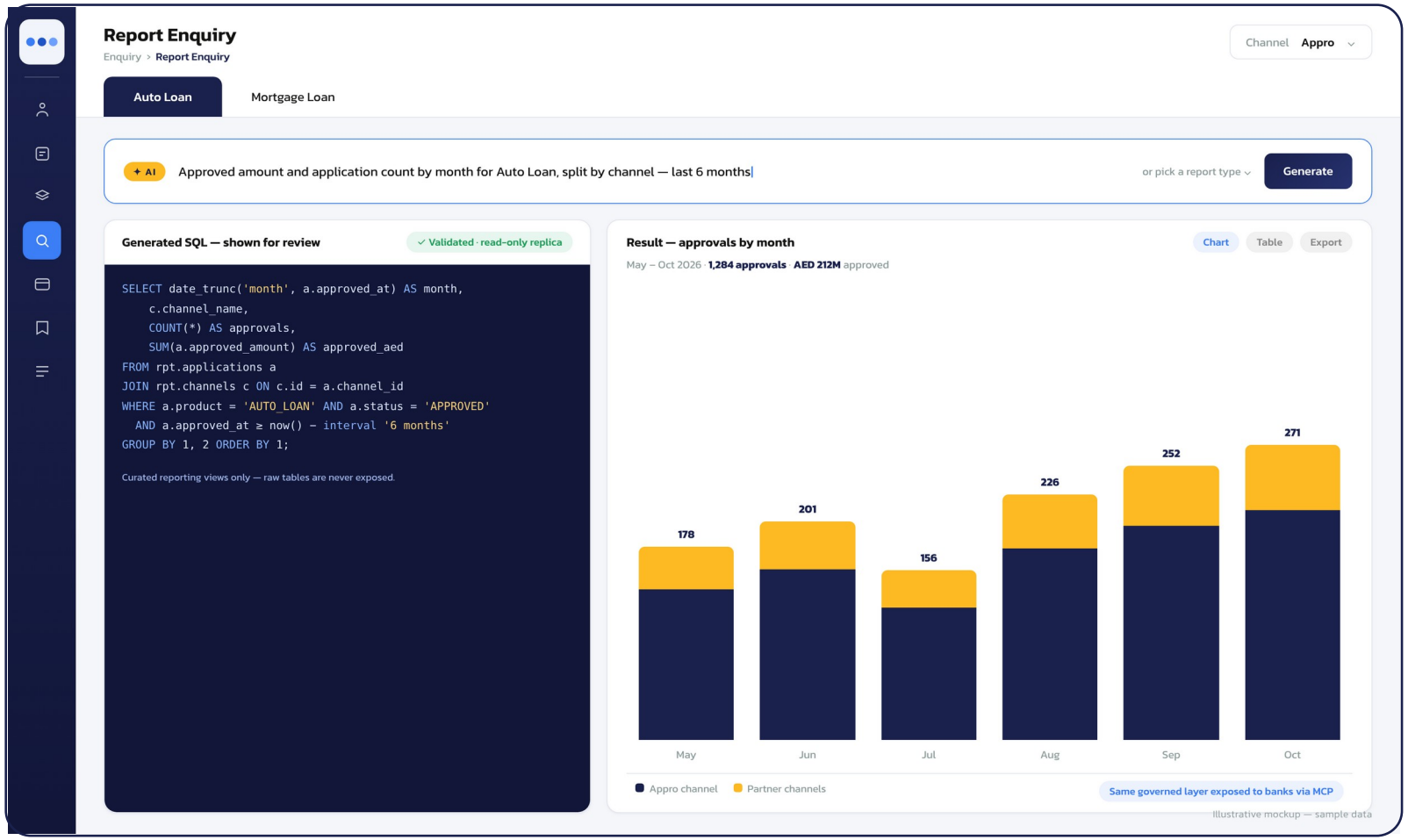
Constrain: curated reporting views + the report types already in Report Enquiry; SQL always visible; read-only replica.

Gate: top report intents validated against analyst-written SQL before free-form prompts unlock. MCP serves the same governed layer — never raw tables.

Sources: BEAVER (arXiv 2409.02038) · Spider 2.0 (ICLR 2025) · BIRD leaderboard (Gemini-SQL2 80.04%)

AI Reports — How It Looks

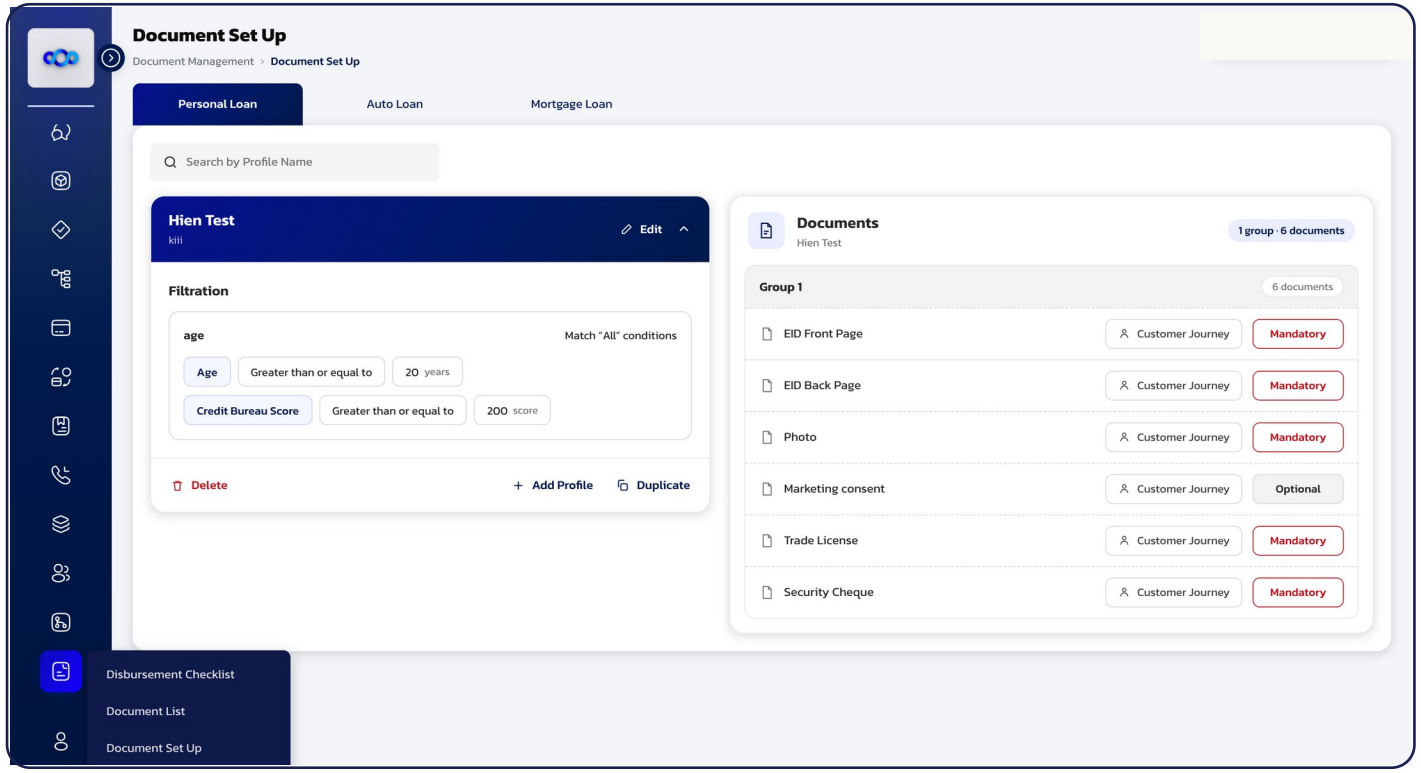
ILLUSTRATIVE MOCKUP — SAMPLE DATA



Report Enquiry with a prompt bar — the generated SQL is always shown for review beside the chart; the same governed reporting views serve bank clients through MCP.

Document Validation — Checkers Review Exceptions, Not Everything

AI pre-checks every document so checkers spend their time on exceptions — not on 100% review.



Document Management › Document Set Up — mandatory documents are already modelled per product and journey.

Requirement · doc comparison (20 Oct)

Validate documents in the new Document setup module — security cheques, source-of-income proofs, clearance letters — against application data before ops review.

Market base

Ocrolus: classifies & analyses 1,150+ financial document types with fraud-signal detection (vendor).

Informed.IQ: automated stip verification; works with 7 of the top-10 US auto lenders (company-reported).

Feasibility & gate

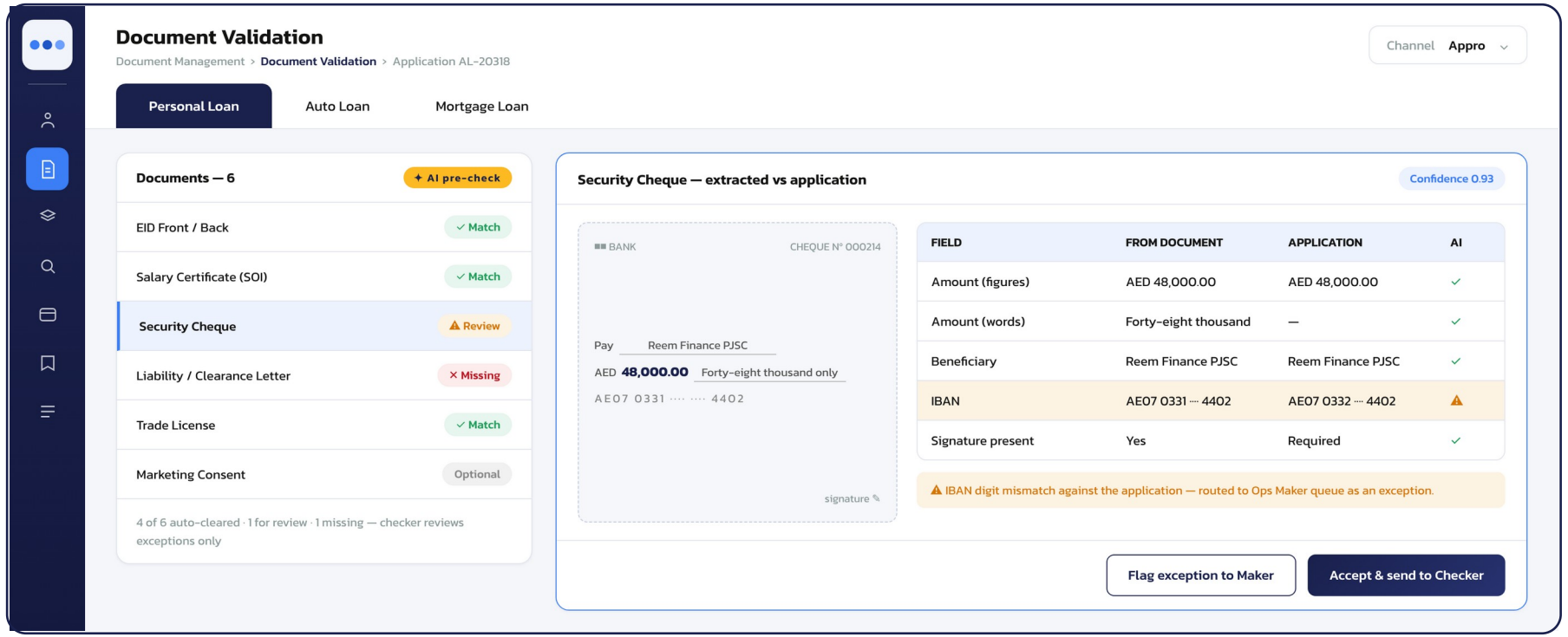
Task: extract fields, compare to the application (amount, names, IBAN, dates), flag missing or inconsistent items into the ops Maker queue.

Dependency: labelled document samples — the largest data gap. Gate: model agrees with checker verdicts on a labelled pilot set.

Sources: Ocrolus (vendor) · Informed.IQ (company-reported)

Document Validation — How It Looks

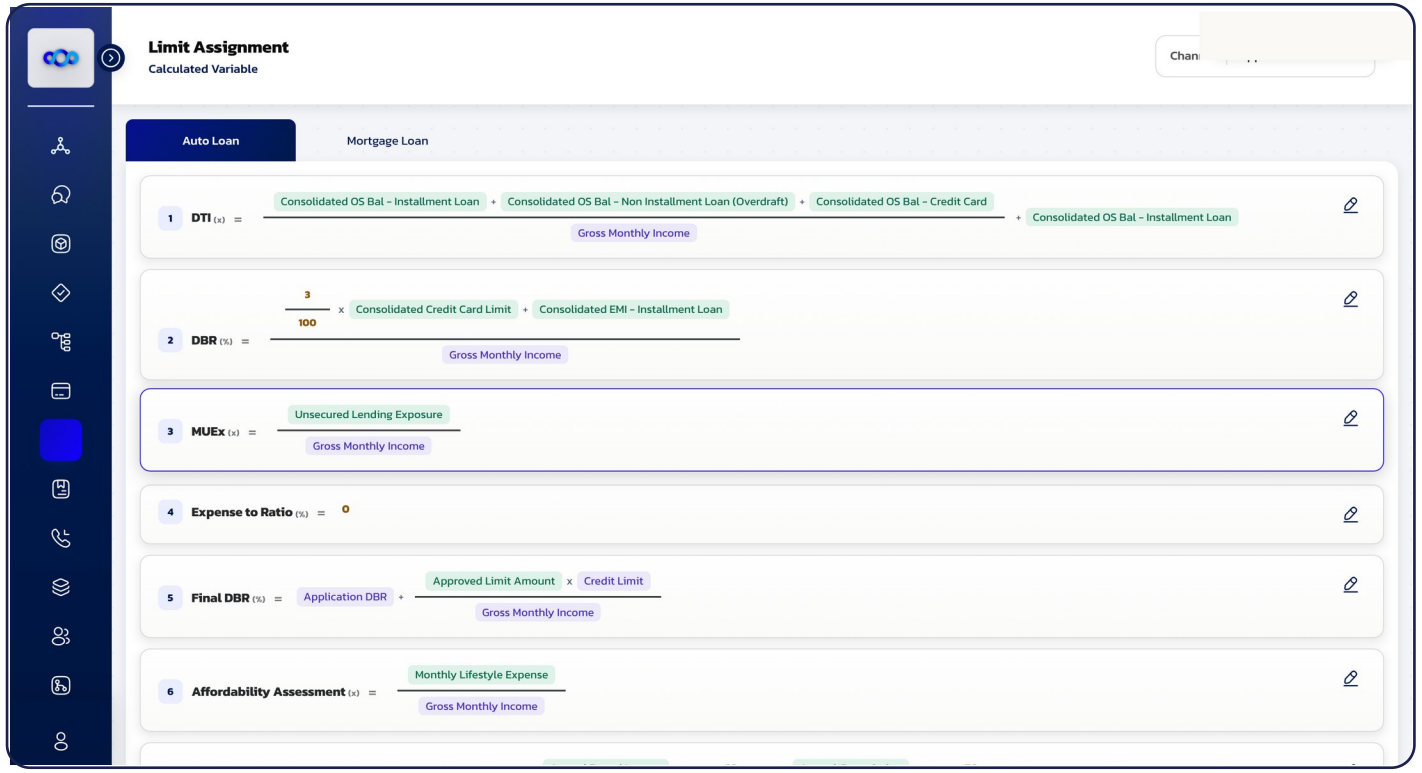
ILLUSTRATIVE MOCKUP — SAMPLE DATA



AI pre-check across the application's documents — fields extracted from the security cheque are compared to the application; matches auto-clear, and only exceptions reach the ops queue.

FEEL Assist — The Fastest Win On The List

The fastest win: FEEL has a formal grammar, a built-in validator — and users who struggle with it today.



Limit Assignment › Calculated Variable — DBR, MUEx, Max Eligibility: the formulas users build by hand today.

Requirement · prompt → FEEL (20 Oct)

Write FEEL formulas from a prompt in Limit Assignment › Calculated Variables — configuring them is too difficult for users today (Anto).

Use case 5 folds in: create new Rule Engine variables with FEEL expressions.

Technical base

FEEL is the DMN-standard expression language — a small formal grammar.

Constrained code generation with instant parser validation is the most mature LLM pattern in production copilots.

Feasibility & gate

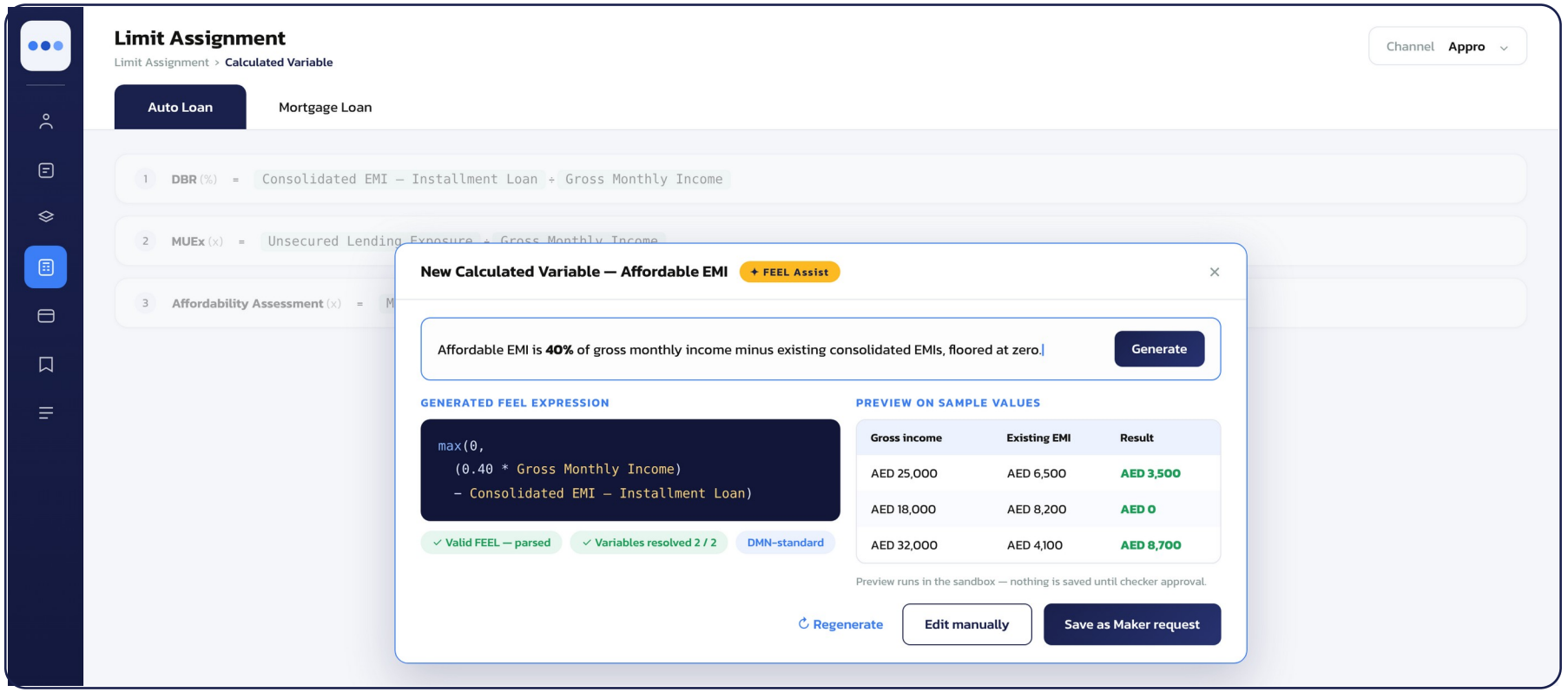
Flow: generate → parse → preview on sample values → save as Maker request.

Zero customer data to start: ground on the existing formula library. Gate: BA panel signs off a replay of the current formulas from plain-language descriptions. Recommended v1.

Sources: OMG DMN / FEEL specification · SMBP portal mirror (Appro build)

FEEL Assist — How It Looks

ILLUSTRATIVE MOCKUP — SAMPLE DATA



FEEL Assist in Limit Assignment › Calculated Variables — plain-language prompt, generated FEEL expression with parser validation, preview on sample values, saved as a Maker request.

Solution Direction — One Model, Four Adapters

One offline small language model behind all four use cases — the adapters differ, the controls stay the same.

Prompt	→	Guardrails	→	SLM	→	Task adapter	→	Validator	→	Maker-Checker
in-module UI		scope + PII filter		offline · UAE-hosted		rule · SQL · doc · FEEL		parser · replay · match		approve & publish

Model direction — Foundry SLM

Shortlist from the Foundry small-model catalog — e.g. the Phi-4 family: reported MIT-licensed and self-hostable (Foundry, Hugging Face, Ollama).

Verify licence and commercials on the model card — Cathal's session output.

One base model, per-task adapters — not four separate models.

Grounding vs fine-tuning

Start with RAG + few-shot on our own artefacts: attribute catalog, schema dictionary, formula library, document profiles.

Fine-tune only where evals prove a gap — most likely prompt→rule and prompt→FEEL.

Each use case keeps its own eval set — the go/no-go instrument per phase.

Reusable if we swap models

Model-agnostic: prompts, task schemas, eval sets, guardrails, validators, UI.

Model-bound: fine-tuned weights only.

Swap the model tomorrow and the training assets survive — the platform investment is protected.

Sources: Azure AI Foundry / Phi model documentation (licence to be confirmed on model card)

Governance — Designed To Pass Review

The CBUAE's Feb-2026 AI guidance maps directly onto maker-checker controls the portal already runs.

CBUAE expectations (Feb 2026)

Guidance Note on responsible AI/ML for licensed financial institutions: human oversight on high-impact decisions, validation before deployment, monitoring for model drift, periodic bias testing. Not legally binding — but the supervisory direction.

Our answer, by design

AI stops at draft. Credit decisions stay deterministic in the Rule Engine; every AI-assisted change is a Maker request with checker approval; new policies run in shadow mode before production.

Auditability

Log per change: prompt, model version, draft, validation result, approver.

Same maker-checker trail banks already audit — no new audit framework to invent.

Data protection

Offline UAE hosting keeps bank data in-country.

Training data anonymised before any use — owned by the data-readiness workstream.

Sources: CBUAE Guidance Note on AI/ML (Feb 2026) — verify specifics against the original text before external use

Data Readiness — What We Have vs What Training Needs

Per Antofelix: Gurkirat + Hailey own data readiness — what we can ground on today, what training needs, and the gaps.

Use case	Grounding — in the platform today	Training / evaluation — needed	Gap → action
Policy Copilot	Rule Engine attribute catalog (~97 filtration attributes), segmentation & strategy configs	Historical applications + decision outcomes for replay; analyst prompt→rule pairs	Anonymised extract of decisioned applications
AI Reports	Report types + schemas behind Report Enquiry	Schema dictionary with business terms; analyst SQL for the top report intents	Build the semantic layer; collect a query corpus
Doc Validation	Document List & Set Up profiles — document types per product and journey	Labelled samples: security cheques, SOI proofs, clearance letters (accepted + rejected)	Largest gap — start collection through ops queues
FEEL Assist	Live formula library (calculated variables per product)	Prompt→FEEL pairs from BA / analyst sessions	Smallest gap — compile from existing configs

This document is classified as INTERNAL & WHITELISTED | Sharing allowed only with approved external domains

Proof Plan — Hypothesis, Experiment, Gate Per Use Case

No feature ships on belief — each one carries a hypothesis, an experiment on our own data, and a pass gate agreed up front.

FEEL Assist

Hypothesis — prompt → FEEL is valid and correct for the formulas our users actually build.

Experiment — replay the existing formula library from plain-language descriptions; score with parser + value preview.

Gate — BA panel signs off accuracy on the replay set.

AI Reports

Hypothesis — on a curated semantic layer, prompt → SQL matches analyst SQL for the top report intents.

Experiment — blind comparison of model vs analyst SQL on the agreed intent list; score result-set equality.

Gate — the agreed intent list passes before free-form prompts unlock.

Policy Copilot

Hypothesis — prompts reproduce intended policy changes without manual rule editing.

Experiment — re-create past policy CRs from their business descriptions; diff against shipped rules; replay the back-test.

Gate — diff-clean on the CR set, then a shadow pilot on one product.

Document Validation

Hypothesis — the model matches checker verdicts on real document sets.

Experiment — run the labelled pilot corpus through extraction + comparison; confusion matrix vs ops decisions.

Gate — checker-agreement threshold set with ops before any rollout.

Delivery Sequence — Ordered By Data Readiness

Sequence by data readiness, not by appeal — each phase ships a working feature and unblocks the next.

PHASE 0
Foundations
model shortlist · UAE hosting · guardrails

→

PHASE 1
FEEL Assist
smallest data gap — proves the pattern

→

PHASE 2
AI Reports
needs the semantic layer

→

PHASE 3
Policy Copilot
needs replay + shadow infra

→

PHASE 4
Doc Validation
needs the labelled corpus

Cathal — solution & model

Leads agenda §3–6: recommended direction, Foundry model shortlist, hosting & running cost, licence and procurement answers.

Gurkirat + Hailey — data readiness

Dataset inventory and gap list per the data-readiness slide: which tables and datasets exist for training and grounding, anonymisation approach, collection plan for the gaps.

Antofelix — objective & tracking

Objective and use-case recap; the session output is a sequenced activity plan — owner and target date per activity, tracked.

Dates: the technical session puts an owner and a target date on every activity — the sequence and dependencies above are what we lock today.

Decision & Next Steps

Asking today: approve the four-feature scope and the delivery sequence — dates follow the technical session.

1 Approve scope & sequence

Four AI features, phased by data readiness (delivery-sequence slide). Use case 5 delivered inside FEEL Assist.

2 Run the technical session

Cathal returns the model shortlist, hosting, cost and licence answers — dates then land on the plan.

3 Start the data-readiness audit

Gurkirat + Hailey deliver the dataset inventory and gap list (data-readiness slide) as a tracked workstream.

4 Spec v1 — FEEL Assist

UX spec for the prompt experience in Limit Assignment › Calculated Variables; the same pattern then reuses across all four modules.

On approval: a clickable walkthrough of the four features inside the portal will be shared as a link.

This document is classified as INTERNAL & WHITELISTED | Sharing allowed only with approved external domains

Thank you

appro

This is not a legally binding document. Highly confidential — not to be shared without written consent.